PRANAY KOCHURI

CRYPTIC CLAWS

THE CRYSTAL OF INFINITE PATTERNS



A MATHEMATICAL ADVENTURE

A Note To The Readers...

A FASCINATING MATHEMATICAL ADVENTURE STORY THAT I THINK YOU'LL ALL ENJOY.

IMAGINE A WORLD WHERE NUMBERS COME TO LIFE AND MATHEMATICAL CONCEPTS ARE NOT JUST ABSTRACT IDEAS BUT THRILLING ADVENTURES WAITING TO BE EXPLORED.

THIS STORY IS NOT JUST ABOUT SOLVING PROBLEMS BUT ALSO ABOUT THE JOY OF DISCOVERY AND THE WONDER OF LEARNING. IT'S A REMINDER THAT MATHEMATICS IS NOT JUST A SUBJECT TO BE STUDIED BUT A WORLD TO BE EXPLORED AND ENJOYED.

I HOPE YOU FIND THIS TALE AS INSPIRING AND CAPTIVATING AS I DO. LET'S EMBRACE THE ADVENTURE AND CONTINUE OUR JOURNEY THROUGH THE MAGICAL WORLD OF MATHEMATICS TOGETHER!







IT WAS A DARK, GRIM, AND STORMY NIGHT. THUNDER CRASHED OMINOUSLY, LIGHTNING FLASHED THROUGH THE WINDOW, ILLUMINATING THE CLUTTERED STUDY OF DETECTIVE DECLAN FROST. RAINDROPS PELTED THE GLASS, CREATING A RHYTHM THAT MATCHED THE DETECTIVE'S RACING THOUGHTS. RAINDROPS PELTED THE GLASS CREATING A RHYTHM THAT MATCHED THE DETECTIVE'S RACING THOUGHTS.

THE ROOM WAS FILLED WITH THE MUSTY SCENT OF OLD BOOKS, WITH A HUGE BOOKSHELF PACKED TIGHT WITH MYSTERY NOVELS. A NOTICEABLE GAP ON THE SHELF REVEALED THE DETECTIVE'S OBSESSION WITH A PARTICULAR SERIES. THE NOVELS LAY SCATTERED ACROSS HIS DESK IN A CHAOTIC MESS, HINTING AT HIS RECENT STRUGGLE TO FOCUS ON ANY SINGLE CASE. JUST AS HE WAS ABOUT TO REACH THE CLIMAX OF HIS CURRENT READ, THE LOUD DOORBELL ECHOED THROUGH THE ROOM, BREAKING THE TENSE ATMOSPHERE.





THE DETECTIVE ROSE FROM HIS CREAKY WOODEN CHAIR, THE SOUND ECHOING OMINOUSLY IN THE DIMLY LIT SPACE. THE APARTMENT DEFINITELY NEEDED SOME REPAIRS. HE SHUFFLED TO THE DOOR AND PEERED THROUGH THE PEEPHOLE. ALL HE COULD SEE WAS THE SILHOUETTE OF A TALL FIGURE CLOAKED IN BLACK.

CURIOSITY PIQUED, HE SLOWLY OPENED THE DOOR, GREETED BY A GUST OF COLD WIND THAT NEARLY PUSHED HIM BACK. THE MYSTERIOUS FIGURE HAD VANISHED!

LEANING OUT TO LOOK DOWN THE CORRIDOR, HE NOTICED THE TALL MAN SLIPPING INTO THE ELEVATOR AT THE END OF THE HALL.





CONFUSED BUT INTRIGUED, THE DETECTIVE SPOTTED A WOODEN BOX LEFT BEHIND BY THE FIGURE. HE PICKED IT UP, FEELING THE WEIGHT OF IT IN HIS HANDS, AND CARRIED IT BACK TO HIS OFFICE, A MIX OF EXCITEMENT AND APPREHENSION BUBBLING WITHIN HIM.

ONCE AT HIS CLUTTERED TABLE, HE EXAMINED THE BOX CLOSELY. IT WAS BEAUTIFULLY CRAFTED, WITH INTRICATE DESIGNS CARVED INTO ITS SURFACE, YET THERE WAS NO SENDER ADDRESS OR NAME--JUST A HEAVY AIR OF MYSTERY.



WITH A TRUCKLOAD OF QUESTIONS SWIRLING IN HIS MIND, THE DETECTIVE RAN HIS FINGERS ALONG THE BOX'S EDGES, SEEKING CLUES. SUDDENLY, HE FOUND A BUTTON CARVED INTO THE WOOD. HE PRESSED IT, AND WITH A WHIRRING SOUND, FOUR PLATES SLID OUT FROM THE SIDES, EACH ADORNED WITH DIFFERENT SHAPE PROBLEMS. HIS EYES LIT UP WITH EXCITEMENT. HE COULDN'T RESIST THE CHALLENGE; SOLVING PUZZLES WAS ONE OF HIS FAVORITE PASTIMES!







A CHEF IS PREPARING A RECIPE AND STARTS WITH 7 CUPS OF MILK. HE USES $2\frac{1}{3}$ FOR A CAKE AND $3\frac{3}{4}$ CUPS FOR A SOUP. HOW MUCH MILK DOES HE HAVE LEFT?:



THIS FIRST PROBLEM REQUIRES US TO LEARN HOW TO WORK WITH MIXED FRACTIONS. A MIXED FRACTION CONTAINS A WHOLE NUMBER AND A FRACTION. TO SIMPLIFY CALCULATIONS, CONVERT IT INTO AN IMPROPER FRACTION BY MULTIPLYING THE WHOLE WITH THE DENOMINATOR AND ADDING THE NUMERATOR. NEXT, WE MUST REMEMBER THAT WHEN ADDING OR SUBTRACTING FRACTIONS, THE DENOMINATORS MUST BE THE SAME. THE EASIEST WAY TO MAKE THE DENOMINATOR COMMON FOR BOTH IMPROPER FRACTIONS IN THE QUESTION IS THE FIND THE LEAST COMMON MULTIPLE OF BOTH DENOMINATORS AND MAKE THE DENOMINATORS THE SAME. WE CAN THEN ADD THESE TO FIND THE VALUE OF THE TOTAL MILK USED AND FINALLY, SUBTRACT THAT VALUE FROM THE 7 CUPS OF MILK TO FIND THE TOTAL AMOUNT OF MILK LEFT.

A SMOOTHIE SHOP MAKES A DRINK USING A SPECIFIC FRUIT RATIO: FOR EVERY 2 CUPS OF STRAWBERRIES, THEY USE 3 CUPS OF BANANAS. IF THE SHOP HAS 12 CUPS OF BANANAS, HOW MANY CUPS OF STRAWBERRIES DO THEY NEED?:



THIS IS A PUZZLE ABOUT RATIO AND PROPORTIONS, WHICH COMPARE TWO QUANTITIES AND HELP US SCALE THEM UP OR DOWN. THE GIVEN RATIO IS 2 CUPS OF STRAWBERRIES FOR EVERY 3 CUPS OF BANANAS, WRITTEN AS 2:3. TO FIND OUT HOW MANY CUPS OF STRAWBERRIES ARE NEEDED WHEN THERE ARE 12 CUPS OF BANANAS, ASSIGN A VARIABLE TO THE VALUE YOU NEED TO FIND. SET UP A PROPORTION IN THE FORM OF TWO EQUIVALENT FRACTIONS AND USE CROSS MULTIPLICATION TO ISOLATE YOUR VARIABLE AND FIND YOUR ANSWER.

A FARMER HAS 12 APPLES AND 18 ORANGES. SHE WANTS TO DIVIDE THEM INTO EQUAL GROUPS SO THAT EACH GROUP HAS THE SAME NUMBER OF APPLES AND ORANGES, WITH NONE LEFT OVER. WHAT IS THE LARGEST NUMBER OF GROUPS HE CAN MAKE?



THIS IS A PUZZLE RELATED TO THE HIGHEST COMMON FACTOR (HCF), WHICH IS THE LARGEST NUMBER THAT DIVIDES TWO NUMBERS EVENLY. TO FIND THE HCF OF TWO NUMBERS, YOU MUST FIRST LIST DOWN THEIR FACTORS. THE FACTORS OF A CERTAIN NUMBER ARE ESSENTIALLY THE NUMBERS THAT CAN BE MULTIPLIED TO GIVE THAT NUMBER. FOR EXAMPLE, 2 AND 5 CAN BE FACTORS OF 10 AS THOSE TWO NUMBERS MULTIPLIED GIVE 10. LIKE THIS, LIST DOWN THE FACTORS OF BOTH THE NUMBERS GIVEN IN THE PROBLEM TO FIND THE HCF, WHICH GIVES THE NUMBER OF GROUPS THAT CAN BE MADE. LASTLY, DIVIDE THE NUMBER OF APPLES AND ORANGES BY THE HCF TO FIND THE EXACT NUMBER OF FRUITS IN EACH GROUP.

AN ADVENTURER IS EXPLORING A CAVE. SHE STARTS AT POSITION +5 ON A NUMBER LINE SHE TAKES 3 STEPS BACK, THEN 7 STEPS FORWARD, AND FINALLY 4 STEPS BACK. WHERE IS SHE NOW?



THE FINAL PUZZLE IN THIS WOODEN BOX IS A SIMPLE ONE ABOUT ADDING AND SUBTRACTING INTEGERS. INTEGERS ARE ESSENTIALLY POSITIVE AND NEGATIVE WHOLE NUMBERS. TO SOLVE THIS QUESTION, YOU MUST START WITH THE ADVENTURER'S POSITION AT +5. FOR ANY STEPS TAKEN BACK, YOU MUST SUBTRACT FROM THE POSITION AND ANY STEPS TAKEN FORWARD MUST BE ADDED TO THE POSITION. SO, YOU FIRST SUBTRACT 3 FROM THE STARTING POSITION TO GET A NEW POSITION, TO WHICH YOU THEN CONTINUE DOING OPERATIONS TO FIND THE FINAL POSITION.:



AFTER ENTERING THE ANSWER TO THE LAST PROBLEM, THE PLATES SLID BACK INTO THE BOX, AND THE GEARS BEGAN RUMBLING OMINOUSLY. THE DETECTIVE PLACED HIS HAND ON THE BOX, FEELING ITS VIBRATIONS, AND THEN HE HEARD A CLICK. STARTLED, HE PULLED BACK HIS HAND, AS IF SHOCKED BY AN ELECTRIC CURRENT.

IN AN INSTANT, THE BOX POPPED OPEN, REVEALING AN ENVELOPE AT ITS CENTER. WITH CAREFUL HANDS, HE PICKED UP THE ENVELOPE AND OPENED IT. INSIDE LAY A BEAUTIFULLY DESIGNED CARD--AN INVITATION TO A PARTY HOSTED BY HIS OLD FRIEND.



FAST FORWARD TO THE DAY OF THE PARTY. THE DETECTIVE EMERGED FROM THE SHOWER, FEELING REFRESHED AND EAGER TO RECONNECT WITH HIS FRIEND. RUMMAGING THROUGH HIS CLOSET, HE FINALLY FOUND THE SHARP SUIT HIS FRIEND HAD GIFTED HIM.



HE DRESSED SWIFTLY, PLACING THE PARTY INVITATION IN HIS POCKET AND GRABBING HIS KEYS. AS HE LOCKED THE DOOR, AN UNSETTLING SENSATION WASHED OVER HIM. HE GLANCED AROUND THE CORRIDOR, SENSING AN INVISIBLE GAZE UPON HIM. SHAKING OFF THE FEELING, HE TOOK THE ELEVATOR DOWN TO THE BASEMENT PARKING GARAGE.





AS HE DROVE THROUGH THE RAIN-SLICKED STREETS, EXCITEMENT BUBBLED WITHIN HIM. THE LIGHTS OF THE CITY TWINKLED LIKE STARS AGAINST THE DARK SKY, AND HE COULD ALMOST HEAR THE LAUGHTER OF FRIENDS WAITING FOR HIM. BUT AS HE MADE HIS WAY TO THE PARTY, THE FEELING OF BEING FOLLOWED CREPT BACK INTO HIS MIND. SUDDENLY, HE GLANCED IN THE REARVIEW MIRROR AND SPOTTED A FIGURE, WITH HIS KEEN EYE, DRESSED IN BLACK, A SHADOW BLENDING WITH THE NIGHT. THE FIGURE WAS A GOOD DISTANCE BEHIND, BUT A SENSE OF URGENCY GRIPPED HIM.





HIS HEART RACED AS HE FOCUSED ON THE ROAD AHEAD. HE TURNED LEFT AT A CORNER, HOPING TO LOSE HIS TAIL. THE STREETS BEGAN TO TWIST AND TURN, AND HE KEPT GLANCING BACK IN THE MIRROR. THE FIGURE REMAINED A STEADY PRESENCE, ALWAYS JUST BEHIND HIM, LIKE A DARK CLOUD FOLLOWING A STORM.





FEELING THE PRESSURE RISE, HE DECIDED TO TAKE A SERIES OF UNEXPECTED TURNS, WEAVING THROUGH SIDE STREETS AND ALLEYWAYS. EACH TIME HE GLANCED BACK, THE FIGURE REMAINED IN PURSUIT, UNDETERRED.





HE SPED UP, TAKING THE NEXT RIGHT TURN, AND THEN ANOTHER. SUDDENLY, HE SPOTTED A CROWDED AREA AHEAD, A BUSTLING PLAZA FILLED WITH PEOPLE. IF HE COULD JUST BLEND IN...

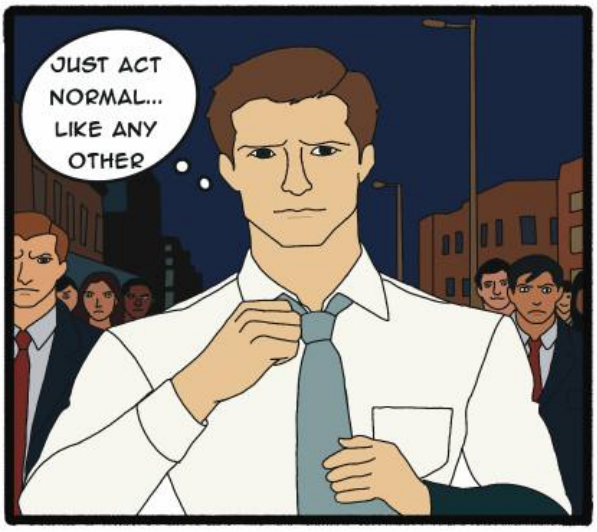
WITH A FINAL SURGE OF SPEED, HE TURNED SHARPLY INTO THE PLAZA AND PARKED. HE QUICKLY GOT OUT, HIS HEART POUNDING, AND BEGAN TO MINGLE WITH THE THRONGS OF PARTYGOERS.



HE SLIPPED OFF HIS JACKET, TOSSING IT OVER HIS ARM, AND BLENDED INTO THE CROWD. AS HE MADE HIS WAY THROUGH THE SEA OF PEOPLE, HE KEPT AN EYE OUT FOR THE SHADOWY FIGURE. AS HE WALKED PAST GROUPS OF LAUGHING FRIENDS AND CHEERFUL CHATTER, HE FINALLY TOOK A DEEP BREATH, FEELING A SENSE OF RELIEF. MAYBE HE'D MANAGED TO SHAKE OFF HIS TAIL AFTER ALL.







AFTER A LARGE HASSLE, THE DETECTIVE HAD ESCAPED FROM HIS TAIL. THOUGHTS BEGAN TO LOOM IN HIS HEAD AS HE MADE HIS WAY THROUGH THE LARGE GROUP OF PARTYGOERS. HE FINALLY MANAGED TO ESCAPE THE CROWD AND WAS MET WITH AN EXTRAORDINARY SITE. A HUGE MEGA MANSION STOOD IN-FRONT OF HIM. HE FOLLOWED THE CROWD OF PARTYGOERS INTO THE MANSION AND WAS MET WITH AN EVEN LARGER WONDER INSIDE. THE LIGHTING, THE DECORATIONS AND ALL THE ACTIVITIES GAVE ILLUMINATION TO THE ROOM.

He walked past many groups of people, laughing and having fun, still slightly paranoid about his unwanted stalker. He calmed himself down by thinking that he did a good job in losing the individual in the crowd earlier. Just as he began to relax, he heard a familiar voice call out to him. It was his friend, waving enthusiastically from across the room.





As the detective exchanged greetings with his friend, Jason, nostalgia washed over him. They talked about their college days, reminiscing over late-night adventures, heated debates, and their shared love of puzzles. Each memory brought laughter, and for a while, the detective let his guard down, enjoying the warmth of friendship amidst the lively party atmosphere.



The night wore on, and the music grew louder. They mingled with other guests, sharing stories, clinking glasses, and marveling at the grandeur of the mansion. But just as the detective felt fully relaxed, he caught a glimpse of something that made his heart skip a beat. Across the room, just for a second, the figure he'd tried to shake off earlier stood, watching him intently from the shadows. His instincts kicked in. He couldn't let this figure keep haunting him





EXCUSING HIMSELF FROM HIS FRIEND, HE MANEUVERED THROUGH THE CROWD, KEEPING HIS EYES LOCKED ON THE DARK FIGURE WHO HAD JUST BEGUN TO SLIP OUT OF SIGHT. AS HE MOVED CLOSER, THE SHADOWY FIGURE SEEMED TO DRIFT TOWARD A HALLWAY LEADING AWAY FROM THE CROWDED MAIN ROOM, DOWN A DARKENED CORRIDOR. HEART POUNDING, HE PICKED UP HIS PACE, TRYING TO CATCH UP WITHOUT DRAWING ATTENTION. EVERY STEP ECHOED THROUGH THE SILENCE AS HE FOLLOWED DEEPER INTO THE SECLUDED WING OF THE MANSION.



THE FIGURE VANISHED AROUND A CORNER, AND THE DETECTIVE QUICKENED HIS PACE, TURNING SHARPLY--ONLY TO FIND HIMSELF IN AN UNFAMILIAR, DESERTED SECTION OF THE BUILDING. THE WALLS WERE LINED WITH OLD PAINTINGS, CASTING EERIE SHADOWS IN THE DIM LIGHT, AND THE SILENCE FELT OPPRESSIVE. JUST AS HE REALIZED HE HAD LOST SIGHT OF THE FIGURE, HE HEARD A QUIET RUSTLING BEHIND HIM. HE WHIPPED AROUND, BUT IT WAS TOO LATE.





SUDDENLY, DARK SHAPES EMERGED FROM HIDDEN ALCOVES AROUND HIM --THREE MENACING FIGURES DRESSED IN BLACK, EACH WITH A LOOK OF STEELY DETERMINATION. BEFORE HE COULD REACT, ONE OF THEM LUNGED FORWARD, PINNING HIS ARMS. HE STRUGGLED, KICKING OUT, BUT ANOTHER FIGURE THREW A PUNCH THAT MADE HIM SEE STARS. DAZED, HE FOUGHT BACK, BUT THEY EASILY OVERPOWERED HIM, DRAGGING HIM DEEPER INTO THE SHADOWS.







ONE FINAL BLOW STRUCK THE SIDE OF HIS HEAD, AND HIS VISION FADED, SLIPPING INTO DARKNESS. AS CONSCIOUSNESS SLIPPED AWAY, HIS LAST THOUGHT WAS A QUESTION THAT HAUNTED HIM: WHO WERE THEY, AND WHAT DID THEY WANT WITH HIM?





THE DETECTIVE JOLTS AWAKE, DISORIENTED. BRIGHT BLUE SKY STRETCHES ENDLESSLY ABOVE HIM, BUT THE GROUND BENEATH FEELS FOREIGN, ALMOST OTHERWORLDLY. HE SITS UP, FEELING A DULL ACHE IN HIS LIMBS, AS IF HE'S BEEN THROWN HERE WITH NO REGARD. AS HE GATHERS HIMSELF, HIS EYES CATCH SIGHT OF AN IMPOSING IRON GATEWAY, TOWERING AND STRANGE. BEYOND IT, A SPRAWLING MANSION LOOMS, DARK AND FOREBODING, WITH IVY-COVERED WALLS THAT SEEM TO MOVE SLIGHTLY, LIKE THEY'RE BREATHING. EVERY INSTINCT TELLS HIM TO STAY AWAY, BUT CURIOSITY PUSHES HIM FORWARD. JUST AS HE STEPS TOWARDS THE GATE, A DEEP, CHILLING VOICE ECHOES AROUND HIM.





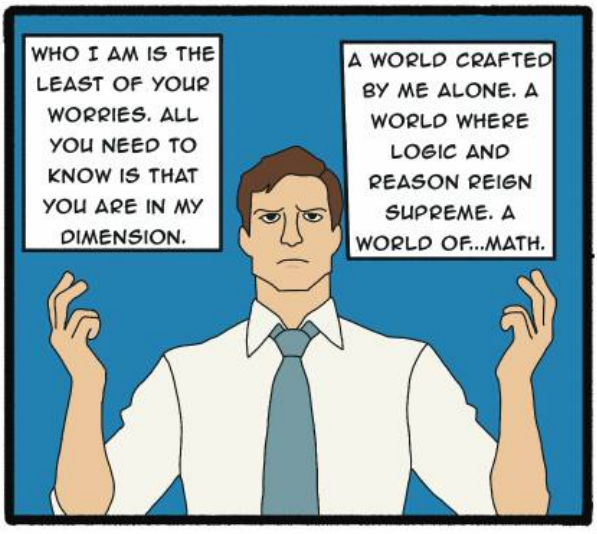


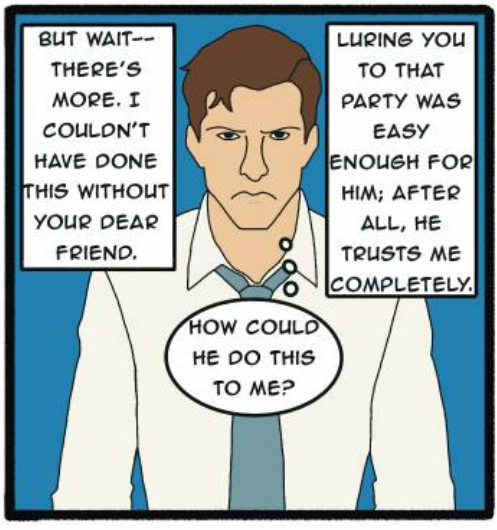




THE DETECTIVE'S MIND RACES. HIS SURROUNDINGS BLUR AS HE TRIES TO PIECE TOGETHER WHERE HE IS AND WHY HE'S BOUND TO THIS BIZARRE DEVICE. PANIC BUBBLES WITHIN, BUT HE FORCES HIMSELF TO STAY CALM, EXAMINING HIS SURROUNDINGS FOR ANY CLUES, ANY WAY OUT.



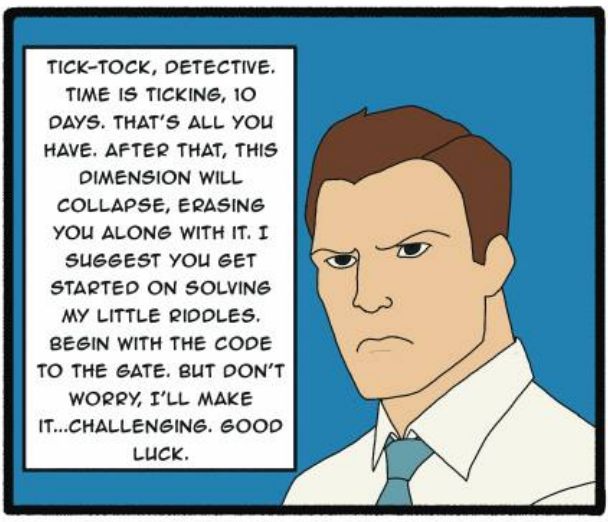




THE SHOCK HITS HIM HARD, BUT ANGER SOON IGNITES IN ITS PLACE. HE CLENCHES HIS FISTS, BARELY RESISTING THE URGE TO BREAK FREE. JUST AS HE STEPS FORWARD TO SLAM HIS FIST AGAINST THE GATE IN FRUSTRATION, THE VOICE RETURNS.



TICK-TOCK, DETECTIVE. TIME IS TICKING, 10 DAYS. THAT'S ALL YOU HAVE. AFTER THAT, THIS DIMENSION WILL COLLAPSE, ERASING YOU ALONG WITH IT. I SUGGEST YOU GET STARTED ON SOLVING MY LITTLE RIDDLES. BEGIN WITH THE CODE TO THE GATE. BUT DON'T WORRY, I'LL MAKE IT...CHALLENGING. GOOD LUCK.





A GARDENER BOUGHT SOME SEEDS. THE TOTAL COST $60, WHICH INCLUDED A $10 DELIVERY FEE. IF EACH PACKET OF SEEDS COSTS $5, HOW MANY PACKETS DID THE GARDENER BUY.



THIS IS A PUZZLE THAT HELPS US UNDERSTAND ALGEBRAIC EXPRESSIONS. THIS PUZZLE INVOLVES SOLVING A TWO-STEP EQUATION. FIRST, ASSIGN A VARIABLE FOR THE UNKNOWN NUMBER OF SEED PACKETS, IN THIS CASE, X. FROM THE QUESTION, WE CAN UNDERSTAND THAT THE TOTAL COST OF THE SEEDS ALSO INCLUDES THE DELIVERY FEE. CREATE AN EQUATION WITH THESE VALUES. THE NEXT THING TO DO WITH THE EQUATION CREATED IS TO REMOVE THE DELIVERY FEES BY SUBTRACTING BOTH SIDES OF THE EQUATION. FINALLY, ISOLATE THE VARIABLE X, AND TO UNDOING THE MULTIPLICATION OF 5X TO GET YOUR ANSWER.



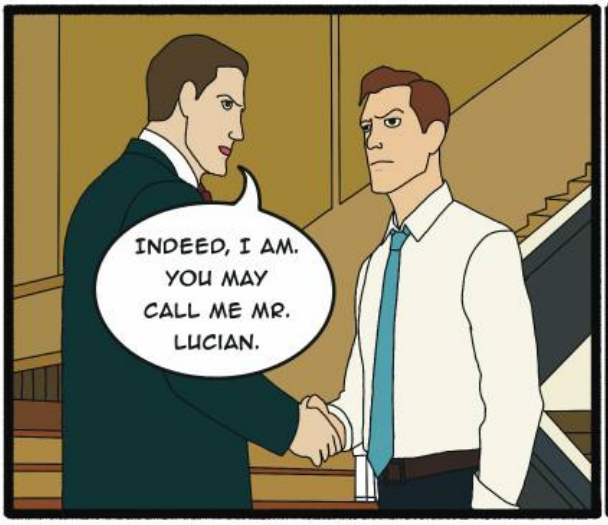
THE GATES CREAK OPEN SLOWLY, REVEALING A LONG, WINDING PATHWAY LINED WITH TOWERING TREES ON EITHER SIDE. THEIR BRANCHES FORM AN ARCH OVERHEAD, CASTING INTRICATE SHADOWS ON THE COBBLESTONE BENEATH. THE DETECTIVE TAKES A DEEP BREATH AND STEPS FORWARD. THE CRUNCH OF GRAVEL UNDERFOOT AND THE FAINT RUSTLE OF LEAVES ARE THE ONLY SOUNDS ACCOMPANYING HIS JOURNEY.

AS HE WALKS, HE CAN'T SHAKE THE SENSE OF UNEASE. THE MANSION LOOMS LARGER WITH EACH STEP--ITS GOTHIC ARCHITECTURE, COMPLETE WITH POINTED ARCHES AND INTRICATE CARVINGS, FEELS BOTH GRAND AND OPPRESSIVE. THE PATH SEEMS LONGER THAN IT SHOULD, BUT EVENTUALLY, HE ARRIVES AT THE MASSIVE DOUBLE DOORS. THE DETECTIVE HESITATES, THEN PUSHES THEM OPEN WITH A GROAN. THE INTERIOR IS DIMLY LIT, WITH FLICKERING CHANDELIERS CASTING A WARM GLOW. A FAINT HUM OF DISTANT CONVERSATION AND THE CLINK OF GLASSES CREATE A STRANGE YET INVITING AMBIANCE.









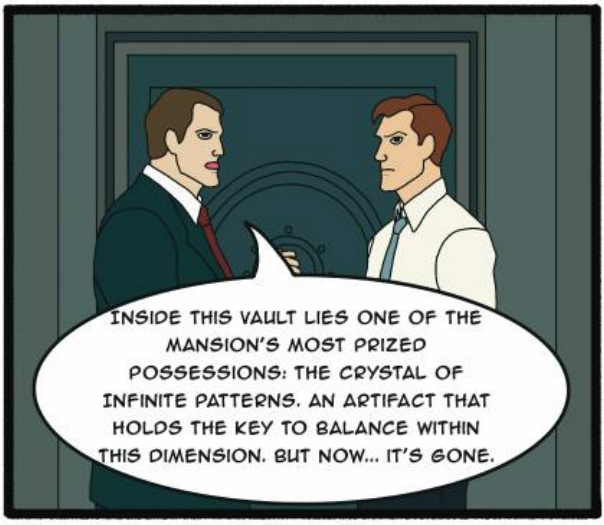
MR. LUCIAN'S HANDSHAKE IS FIRM, HIS SMILE POLITE YET UNNERVING. HE LEADS THE DETECTIVE DEEPER INTO THE MANSION, THE ORNATE DECOR HINTING AT WEALTH AND SECRETS. THEY STOP IN A LAVISH SITTING ROOM, WHERE MR. LUCIAN GESTURES FOR THE DETECTIVE TO SIT. THE DETECTIVE LISTENS INTENTLY, STILL PROCESSING THE SITUATION. THE OWNER OF THE MANSION, HIS VOICE CALM YET SINISTER, DROPS THE BOMBSHELL: A ROBBERY HAS TAKEN PLACE INSIDE THE MANSION, AND IT'S UP TO THE DETECTIVE TO SOLVE IT.







THE DETECTIVE'S MIND RACES, BUT HE KEEPS HIS COMPOSURE. HE KNOWS HE HAS NO CHOICE BUT TO PLAY ALONG. THE OWNER GESTURES FOR HIM TO FOLLOW, LEADING HIM THROUGH A GRAND CORRIDOR LINED WITH EERIE PORTRAITS AND STATUES. EACH STEP ECHOES OMINOUSLY, AMPLIFYING THE TENSION. FINALLY, THEY ARRIVE AT A HEAVILY SECURED ROOM.









THE DETECTIVE REALIZES THE URGENCY OF THE TASK. EVERY SECOND COUNTS, NOT JUST TO RECOVER THE ARTIFACT BUT ALSO TO PREVENT THE DIMENSION FROM COLLAPSING. HE DECIDES TO START BY INTERROGATING THE MANSION'S RESIDENTS AND ANALYZING THE PUZZLES THEY CONTROL, HOPING THE TRAIL WILL LEAD HIM TO THE MISSING CRYSTAL OF INFINITE PATTERNS. THE DETECTIVE DECIDES HIS FIRST STEP IS TO GATHER INFORMATION ABOUT THE MANSION'S RESIDENTS. HE APPROACHES MR. LUCIAN. THE FIRE CRACKLES SOFTLY, CASTING FLICKERING SHADOWS ON THE WALLS.













THE DETECTIVE BEGINS HIS INVESTIGATION. EACH RESIDENT'S ROOM IS LOCKED BY A UNIQUE PUZZLE, A CHALLENGE DESIGNED TO TEST INTELLECT AND CREATIVITY. AS HE TRAVERSES THE MANSION, ITS STRANGE ARCHITECTURE STARTS TO UNSETTLE HIM--A COMBINATION OF ELEGANCE AND EERIE DECAY. ON THE WAY TO THE FIRST SUSPECT, HE NOTICES A CORRIDOR ENTIRELY SHROUDED IN DARKNESS, ITS PRESENCE JARRING AMIDST THE BRIGHTLY LIT HALLS.





THE DETECTIVE STEPS AWAY FROM THE DARK CORRIDOR, DETERMINED TO START HIS INVESTIGATION WITH THE FIRST SUSPECT: DR. ELEANOR WREN, THE SCHOLAR. HE FOLLOWS THE DIRECTIONS GIVEN TO HIM, FINDING A STURDY WOODEN DOOR MARKED WITH INTRICATE CARVINGS. A PLAQUE READS: "KNOWLEDGE IS THE KEY TO EVERY DOOR." A MATH PROBLEM WAS PRESENT ON THE DOOR, WHICH WOULD OPEN IT.



THE FOLLOWING ARE THE TEST SCORES OF 5 STUDENTS: 85, 92, 78, 88, 95. WHAT IS THE MEAN SCORE?

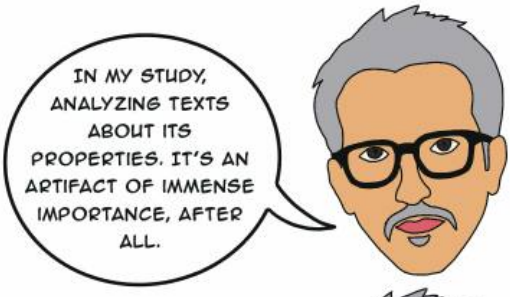


TO SOLVE THIS PROBLEM, WE NEED TO FIND THE MEAN, OR AVERAGE, OF THE TEST SCORES. FIRST, ADD ALL THE TEST SCORES TOGETHER. THIS WILL GIVE YOU THE TOTAL SUM OF THE SCORES. THEN, COUNT HOW MANY TEST SCORES THERE ARE (IN THIS CASE, 5). TO FIND THE MEAN, DIVIDE THE TOTAL SUM BY THE NUMBER OF TEST SCORES. THIS WILL GIVE YOU THE AVERAGE, WHICH REPRESENTS A TYPICAL SCORE FROM THE GROUP.

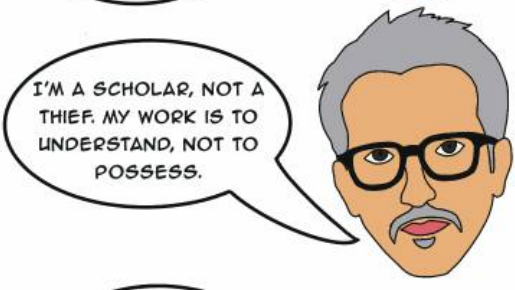

I SEE YOU'RE BUSY. MIND IF I INTERRUPT?


DETECTIVE DECLAN FROST. TO WHAT DO I OWE THIS PLEASURE?


JUST A FEW QUESTIONS. I'M SURE YOU WON'T MIND. LET'S START SIMPLE: WHERE WERE YOU WHEN THE CRYSTAL WENT MISSING?


IN MY STUDY, ANALYZING TEXTS ABOUT ITS PROPERTIES. IT'S AN ARTIFACT OF IMMENSE IMPORTANCE, AFTER ALL.

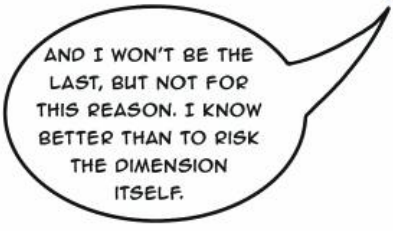

FASCINATION CAN BE A DANGEROUS THING. ARE YOU SURE IT STAYED THEORETICAL?


I'M A SCHOLAR, NOT A THIEF. MY WORK IS TO UNDERSTAND, NOT TO POSSESS.


AND YET, THESE TEXTS YOU HAVE HERE SEEM TO FOCUS ON HOW TO MANIPULATE ITS ENERGY. CARE TO COMMENT?


IT'S RESEARCH. THE CRYSTAL IS UNPREDICTABLE, AND KNOWING HOW TO STABILIZE IT COULD SAVE LIVES.


OR CONTROL LIVES. YOU WOULDN'T BE THE FIRST TO CROSS THAT LINE, DR. WREN.


AND I WON'T BE THE LAST, BUT NOT FOR THIS REASON. I KNOW BETTER THAN TO RISK THE DIMENSION ITSELF.

THE DETECTIVE LEAVES THE SCHOLAR'S QUARTERS AND MOVES TOWARD THE INVENTOR'S WORKSHOP, GUIDED BY THE RHYTHMIC CLANKING OF TOOLS ON METAL. THE WORKSHOP'S DOOR, A SLEEK METALLIC DESIGN ADORNED WITH GEARS AND LEVERS, BEARS A SIGN: "INNOVATION IS BORN FROM CHAOS." THIS ROOM, HE SUSPECTS, MAY HOLD MORE SECRETS THAN ANSWERS.



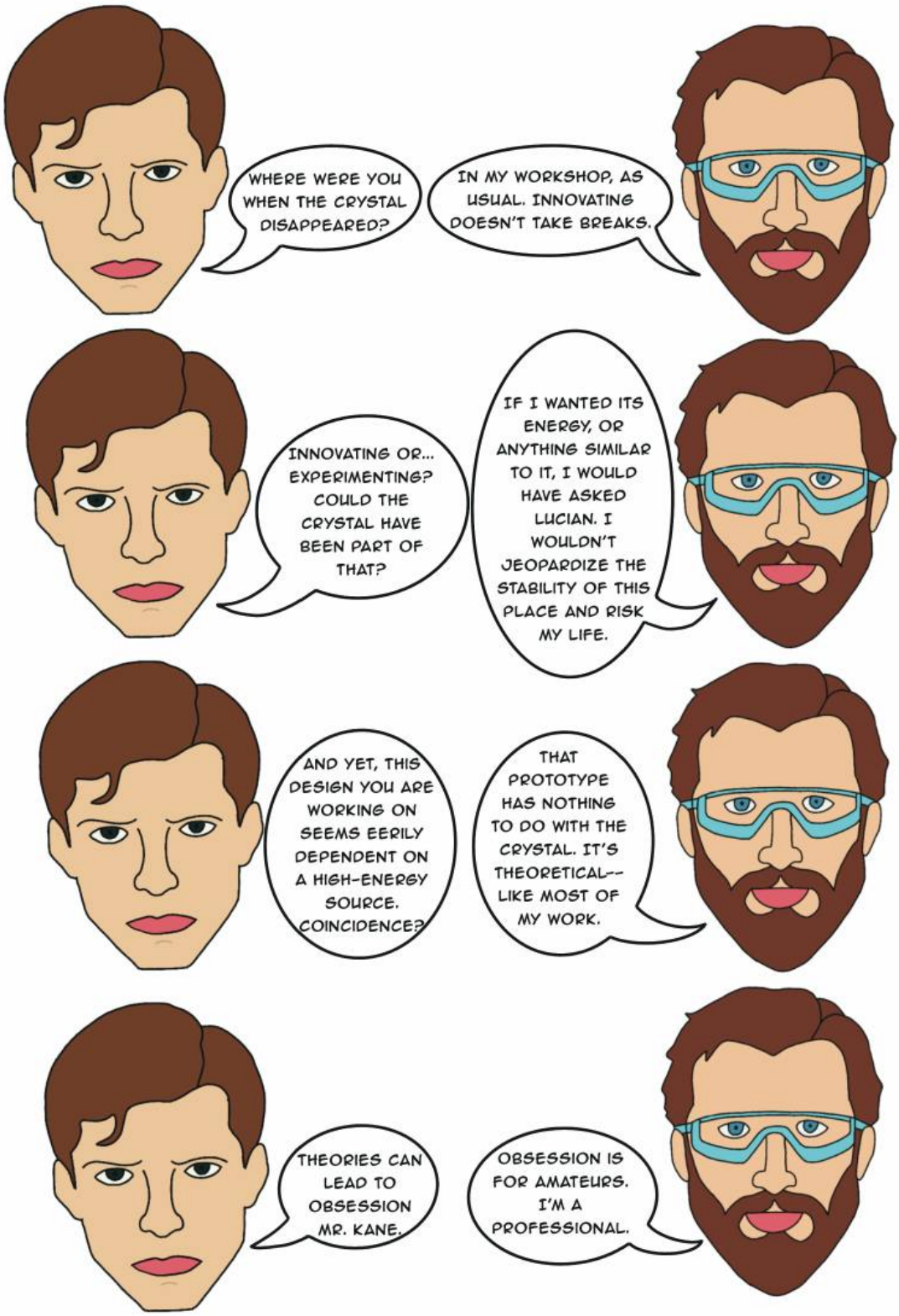
A BAG CONTAINS 6 RED MARBLES, 8 BLUE MARBLES, AND 4 YELLOW MARBLES. WHAT IS THE PROBABILITY OF DRAWING A YELLOW MARBLE?



THIS PROBLEM IS ABOUT FINDING THE PROBABILITY OF AN EVENT HAPPENING. FIRST, ADD THE TOTAL NUMBER OF MARBLES TOGETHER TO GET THE TOTAL NUMBER OF POSSIBLE OUTCOMES. NEXT, FOCUS ON THE NUMBER OF YELLOW MARBLES SINCE WE'RE INTERESTED IN DRAWING ONE. THE PROBABILITY IS THE RATIO OF THE NUMBER OF YELLOW MARBLES TO THE TOTAL NUMBER OF MARBLES. TO SOLVE THIS, DIVIDE THE NUMBER OF YELLOW MARBLES BY THE TOTAL NUMBER OF MARBLES TO FIND THE LIKELIHOOD OF DRAWING A YELLOW MARBLE.






WHERE WERE YOU WHEN THE CRYSTAL DISAPPEARED?
IN MY WORKSHOP, AS USUAL. INNOVATING DOESN'T TAKE BREAKS.
INNOVATING OR... EXPERIMENTING? COULD THE CRYSTAL HAVE BEEN PART OF THAT?
IF I WANTED ITS ENERGY, OR ANYTHING SIMILAR TO IT, I WOULD HAVE ASKED LUCIAN. I WOULDN'T JEOPARDIZE THE STABILITY OF THIS PLACE AND RISK MY LIFE.
AND YET, THIS DESIGN YOU ARE WORKING ON SEEMS EERILY DEPENDENT ON A HIGH-ENERGY SOURCE. COINCIDENCE?
THAT PROTOTYPE HAS NOTHING TO DO WITH THE CRYSTAL. IT'S THEORETICAL-- LIKE MOST OF MY WORK.
THEORIES CAN LEAD TO OBSESSION MR. KANE.
OBSESSION IS FOR AMATEURS. I'M A PROFESSIONAL.

FROM THE WORKSHOP, THE DETECTIVE MAKES HIS WAY TO THE MANSION'S GRAND LIBRARY, WHERE SOPHIA DUVALL RESIDES. THE MASSIVE OAK DOORS ARE ETCHED WITH SCENES OF ANCIENT CIVILIZATIONS, THEIR ACHIEVEMENTS FROZEN IN TIME. ABOVE THE DOORWAY, AN INSCRIPTION READS: "THE PAST IS A GUIDE TO THE FUTURE." HE TAKES A LOOK AT THE PROBLEM ON THE DOOR



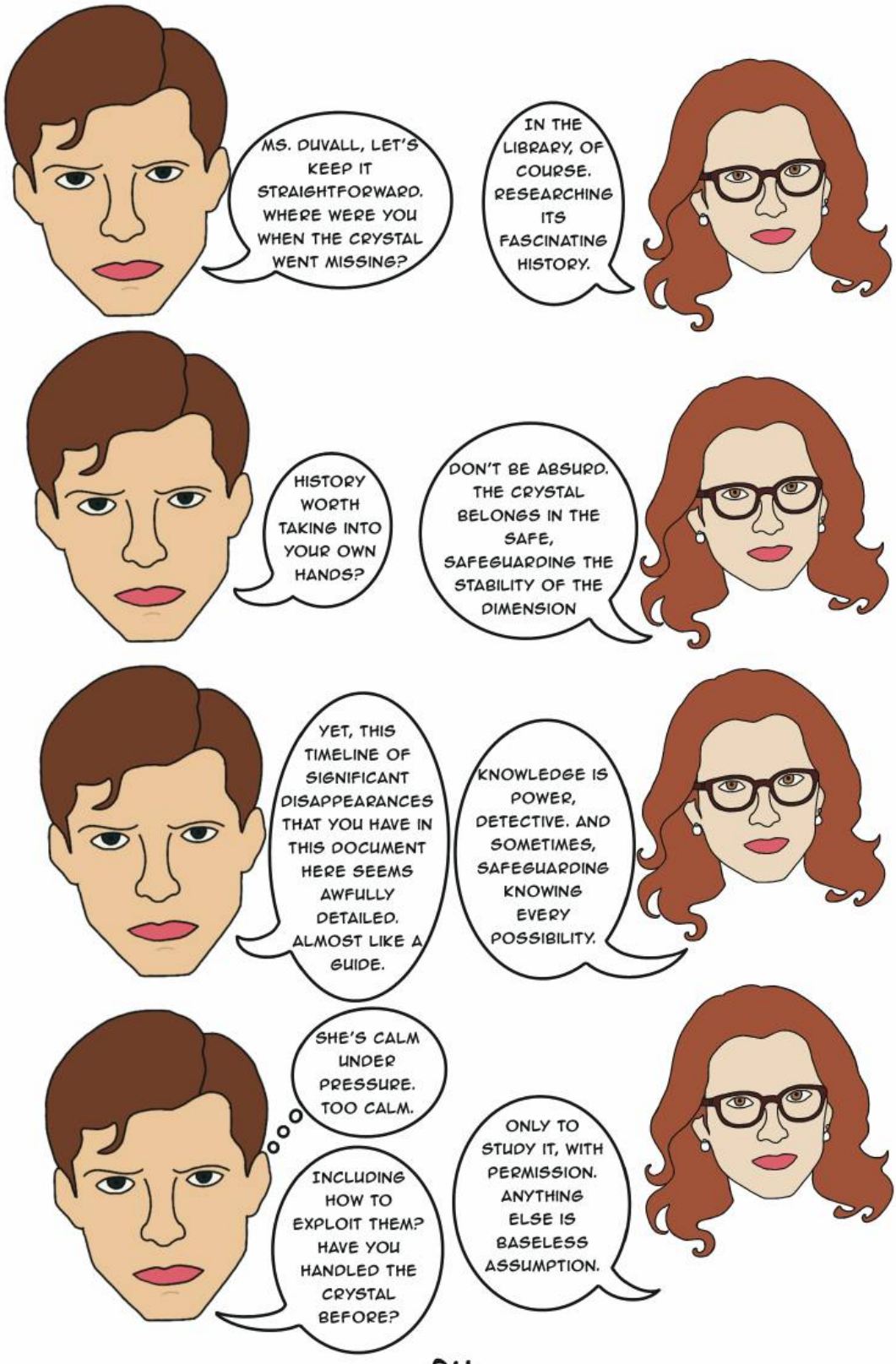
A RECTANGULAR PRISM HAS A LENGTH OF 6 CM, A WIDTH OF 4 CM, AND A HEIGHT OF 3 CM. WHAT IS THE VOLUME OF THE PRISM?



TO SOLVE THIS PROBLEM, WE NEED TO FIND THE VOLUME OF A RECTANGULAR PRISM. VOLUME MEASURES HOW MUCH SPACE A 3D OBJECT TAKES UP. FOR A RECTANGULAR PRISM, THE VOLUME IS FOUND BY MULTIPLYING THE THREE DIMENSIONS: LENGTH, WIDTH, AND HEIGHT. START BY MULTIPLYING THE LENGTH AND WIDTH, AND THEN MULTIPLY THE RESULT BY THE HEIGHT. THIS WILL GIVE YOU THE TOTAL VOLUME OF THE PRISM, REPRESENTING THE SPACE INSIDE IT.






MS. DUVALL, LET'S KEEP IT STRAIGHTFORWARD. WHERE WERE YOU WHEN THE CRYSTAL WENT MISSING?
IN THE LIBRARY, OF COURSE. RESEARCHING ITS FASCINATING HISTORY.
HISTORY WORTH TAKING INTO YOUR OWN HANDS?
DON'T BE ABSURD. THE CRYSTAL BELONGS IN THE SAFE, SAFEGUARDING THE STABILITY OF THE DIMENSION
YET, THIS TIMELINE OF SIGNIFICANT DISAPPEARANCES THAT YOU HAVE IN THIS DOCUMENT HERE SEEMS AWFULLY DETAILED. ALMOST LIKE A GUIDE.
KNOWLEDGE IS POWER, DETECTIVE. AND SOMETIMES, SAFEGUARDING KNOWING EVERY POSSIBILITY.
SHE'S CALM UNDER PRESSURE. TOO CALM.
INCLUDING HOW TO EXPLOIT THEM? HAVE YOU HANDLED THE CRYSTAL BEFORE?
ONLY TO STUDY IT, WITH PERMISSION. ANYTHING ELSE IS BASELESS ASSUMPTION.

THE DETECTIVE ASCENDS TO THE UPPER FLOORS, WHERE ISABELLE CROWLEY'S OFFICE IS TUCKED AWAY IN A SECLUDED WING. THE CORRIDOR LEADING TO HER OFFICE IS LINED WITH PORTRAITS OF PAST PROPRIETORS, THEIR AUSTERE GAZES FIXED ON THE DETECTIVE AS HE APPROACHES. THE DOOR TO HER OFFICE IS UNASSUMING BUT METICULOUSLY POLISHED, WITH A BRASS PLATE THAT READS: "CONTROL THE RESOURCES, CONTROL THE WORLD. A SMALL SCREEN IS PLACED BENEATH THE SIGN, REVEALING A MATH PROBLEM.



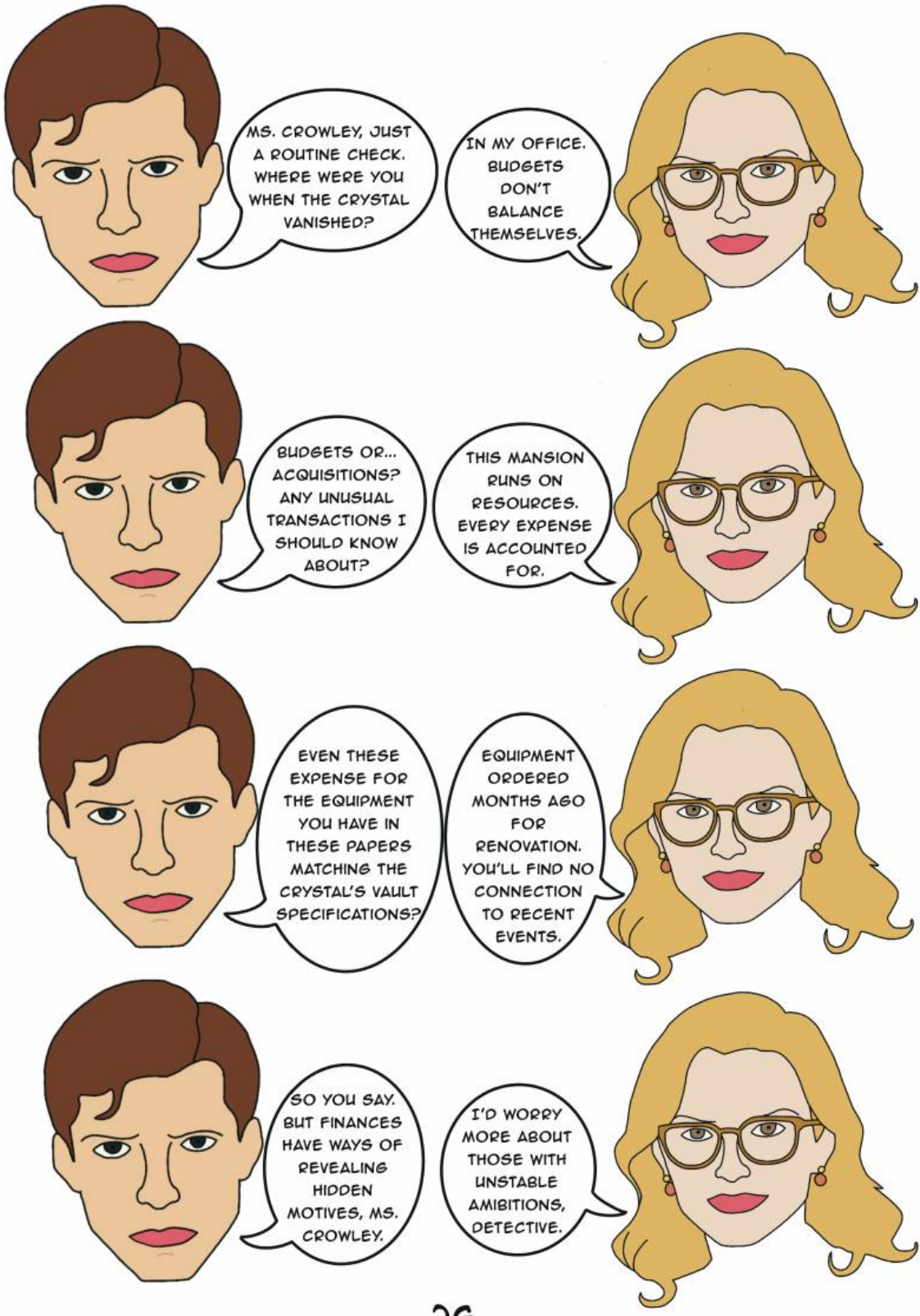
YOU HAVE A ROPE THAT IS 5,000 CENTIMETERS LONG. YOU NEED TO CUT IT INTO PIECES, EACH 2.5 METERS LONG. HOW MANY PIECES CAN YOU CUT FROM THE ROPE?



THIS PROBLEM REQUIRES CONVERTING CENTIMETERS TO METERS BEFORE PERFORMING DIVISION. SINCE 100 CENTIMETERS EQUALS 1 METER, WE START BY CONVERTING THE TOTAL LENGTH OF THE ROPE INTO METERS BY DIVIDING 5,000 BY 100. ONCE WE HAVE THE TOTAL LENGTH IN METERS, WE CAN DIVIDE IT BY THE LENGTH OF EACH PIECE (2.5 METERS) TO FIND THE NUMBER OF PIECES WE CAN CUT. THIS TEACHES US HOW TO USE MEASUREMENT CONVERSIONS AND DIVISION TO SOLVE REAL-WORLD PROBLEMS.






MS. CROWLEY, JUST A ROUTINE CHECK. WHERE WERE YOU WHEN THE CRYSTAL VANISHED?
IN MY OFFICE. BUDGETS DON'T BALANCE THEMSELVES.
BUDGETS OR... ACQUISITIONS? ANY UNUSUAL TRANSACTIONS I SHOULD KNOW ABOUT?
THIS MANSION RUNS ON RESOURCES. EVERY EXPENSE IS ACCOUNTED FOR.
EVEN THESE EXPENSE FOR THE EQUIPMENT YOU HAVE IN THESE PAPERS MATCHING THE CRYSTAL'S VAULT SPECIFICATIONS?
EQUIPMENT ORDERED MONTHS AGO FOR RENOVATION. YOU'LL FIND NO CONNECTION TO RECENT EVENTS.
SO YOU SAY. BUT FINANCES HAVE WAYS OF REVEALING HIDDEN MOTIVES, MS. CROWLEY.
I'D WORRY MORE ABOUT THOSE WITH UNSTABLE AMIBITIONS, DETECTIVE.

LASTLY, THE DETECTIVE VENTURES TO THE GUARD QUARTERS IN THE MANSION'S LOWER LEVEL, A STARK CONTRAST TO THE OPULENCE ABOVE. THE WALLS ARE BARE, THE LIGHTING DIM, AND THE AIR HEAVY WITH THE SCENT OF DAMP STONE. THE DOOR TO MARCUS HALE'S QUARTERS IS REINFORCED WITH STEEL, REFLECTING HIS NO-NONSENSE DEMEANOR. A SIMPLE PLACARD READS: "STRENGTH IS LOYALTY." AS THE DETECTIVE IS ABOUT TO TAKE A LOOK AT THE PROBLEM ON THE DOOR, HE NOTICES SCUFF MARKS ON THE FLOOR OUTSIDE THE DOOR--SUBTLE SIGNS OF ACTIVITY, BUT FROM WHAT OR WHOM?



YOU ARE DESIGNING A GARDEN IN THE SHAPE OF A TRAPEZOID. THE LENGTHS OF THE TWO PARALLEL SIDES ARE 10 METERS AND 6 METERS, AND THE HEIGHT BETWEEN THEM IS 8 METERS. WHAT IS THE AREA OF THE GARDEN?

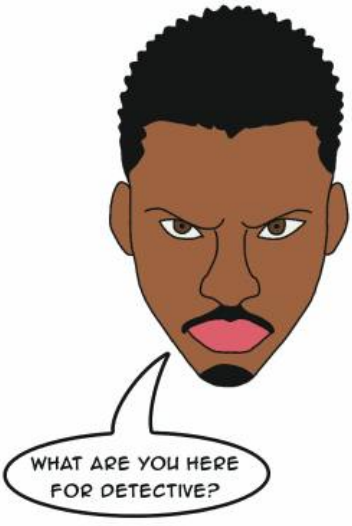


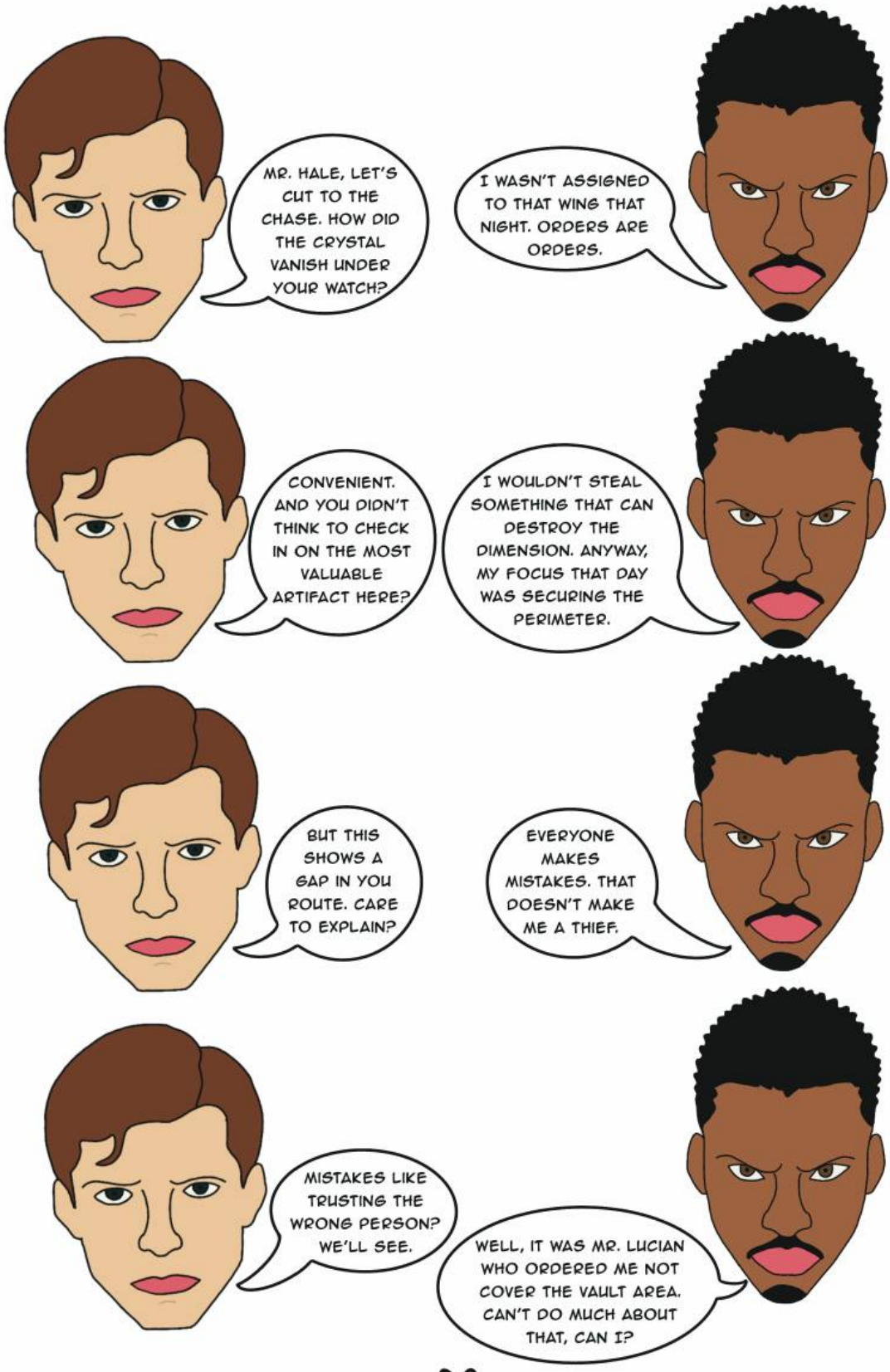
TO SOLVE THIS PROBLEM, WE NEED TO USE THE FORMULA FOR THE AREA OF A TRAPEZOID, WHICH IS:

$$\text{AREA} = \frac{1}{2} \times (\text{BASE1} + \text{BASE2}) \times \text{HEIGHT}$$

START BY ADDING THE LENGTHS OF THE TWO PARALLEL SIDES (THE BASES). THEN MULTIPLY THE SUM BY THE HEIGHT AND DIVIDE BY 2 TO FIND THE AREA. THIS TEACHES US HOW TO WORK WITH SHAPES THAT AREN'T REGULAR RECTANGLES OR SQUARES, USING SPECIFIC FORMULAS TO FIND THEIR AREA.






MR. HALE, LET'S CUT TO THE CHASE. HOW DID THE CRYSTAL VANISH UNDER YOUR WATCH?
I WASN'T ASSIGNED TO THAT WING THAT NIGHT. ORDERS ARE ORDERS.
CONVENIENT. AND YOU DIDN'T THINK TO CHECK IN ON THE MOST VALUABLE ARTIFACT HERE?
I WOULDN'T STEAL SOMETHING THAT CAN DESTROY THE DIMENSION. ANYWAY, MY FOCUS THAT DAY WAS SECURING THE PERIMETER.
BUT THIS SHOWS A GAP IN YOU ROUTE. CARE TO EXPLAIN?
EVERYONE MAKES MISTAKES. THAT DOESN'T MAKE ME A THIEF.
MISTAKES LIKE TRUSTING THE WRONG PERSON? WE'LL SEE.
WELL, IT WAS MR. LUCIAN WHO ORDERED ME NOT COVER THE VAULT AREA. CAN'T DO MUCH ABOUT THAT, CAN I?

AS THE DETECTIVE WRAPPED UP HIS FINAL INTERROGATION, HE STEPPED INTO THE HALLWAY, HIS THOUGHTS PREOCCUPIED WITH CONNECTING THE PIECES OF THE PUZZLE. THAT'S WHEN HE SPOTTED MR. LUCIAN AT THE FAR END, GLANCING AROUND NERVOUSLY BEFORE SLIPPING INTO HIS ROOM. AFTER ENSURING THE HALLWAY WAS CLEAR, THE DETECTIVE APPROACHED LUCIAN'S DOOR. HE PRESSED HIS EAR TO THE SURFACE, CATCHING FAINT SOUNDS OF HURRIED MOVEMENT INSIDE. DETERMINED TO UNCOVER THE TRUTH, HE LOOKED AT THE SCREEN WHICH HAD A MATH PROBLEM TO OPEN THE DOOR



A STORE IS OFFERING A 25% DISCOUNT ON A JACKET THAT ORIGINALLY COSTS $80. IF YOU HAVE AN ADDITIONAL COUPON THAT TAKES 10% OFF THE DISCOUNTED PRICE, HOW MUCH WILL YOU END UP PAYING FOR THE JACKET?



THIS PROBLEM INVOLVES WORKING WITH PERCENTAGES IN TWO STEPS. FIRST, WE CALCULATE 25% OF THE ORIGINAL PRICE TO FIND OUT HOW MUCH THE JACKET IS DISCOUNTED BY. SUBTRACT THIS DISCOUNT FROM THE ORIGINAL PRICE TO GET THE DISCOUNTED PRICE. NEXT, WE APPLY THE 10% COUPON ON THE DISCOUNTED PRICE BY CALCULATING 10% OF THAT AMOUNT AND SUBTRACTING IT. THIS TEACHES US HOW TO APPLY SUCCESSIVE PERCENTAGES, ONE AFTER THE OTHER, TO SOLVE REAL-WORLD PROBLEMS.

INSIDE THE ROOM, LUCIAN STOOD NEAR A DESK, HIS BACK TO THE DOOR. HIS HANDS MOVED FRANTICALLY OVER A BLACK CLOTH, REVEALING GLIMPSES OF A MYSTERIOUS OBJECT UNDERNEATH. THE ROOM WAS CLUTTERED WITH PAPERS, ARTIFACTS, AND BOOKS STACKED RANDOMLY. DECIDING NOT TO DRAW ATTENTION, THE DETECTIVE SLIPS BACK OUT OF THE ROOM, LOCKING THE DOOR BEHIND HIM. HE WOULD NEED MORE EVIDENCE BEFORE ACCUSING LUCIAN DIRECTLY.

AFTER A LONG DAY OF INVESTIGATING, THE DETECTIVE MADE HIS WAY BACK TO THE LIVING QUARTERS MR. LUCIAN HAD PROVIDED FOR HIM. HE QUICKLY FRESHENED UP AND CHANGED INTO A FRESH SET OF CLOTHES FROM THE CLOSET. EXHAUSTED, HE HAD DINNER AND RETURNED TO LIE DOWN ON THE BED. THOUGH THE BED WAS COMFORTABLE, HIS MIND KEPT RACING WITH THOUGHTS OF THE DAY'S EVENTS.



WITH THE FIRST DAY OF INVESTIGATION COMPLETE, THE DETECTIVE ROSE EARLY ON THE SECOND DAY, INTENT ON KEEPING A CLOSE WATCH ON EACH SUSPECT. THE ATMOSPHERE OF THE MANSION FELT DIFFERENT IN THE DAYLIGHT, LESS EERIE BUT NO LESS SUSPICIOUS. EVERY ROOM HELD SECRETS, AND THE PEOPLE INHABITING THEM WERE EVEN MORE ENIGMATIC. HE KNEW THAT SUBTLE OBSERVATION COULD REVEAL WHAT DIRECT INTERROGATION COULD NOT.

OVER THE COURSE OF THE DAY, HE ROAMED THE HALLWAYS AND LINGERED IN COMMON AREAS, NOTING EVERY MOVEMENT, EVERY GLANCE EXCHANGED BETWEEN THE MANSION'S INHABITANTS. DR. WREN IMMERSED HERSELF IN HER STUDIES, RARELY EMERGING FROM HER QUARTERS. VICTOR KANE TINKERED AWAY IN HIS WORKSHOP, THE SOUND OF CLANGING METAL ECHOING THROUGH THE MANSION'S HALLS. SOPHIA DUVALL SPENT MOST OF HER TIME IN THE GRAND LIBRARY, PORING OVER ANCIENT TEXTS. ISABELLE CROWLEY MOVED BRISKLY BETWEEN HER OFFICE AND THE CENTRAL HALL, ALWAYS WITH DOCUMENTS IN HAND, AND MARCUS HALE PATROLLED THE MANSION DILIGENTLY, HIS EYES SHARP AND WATCHFUL.

LUCIAN, HOWEVER, KEPT MOSTLY TO HIMSELF, EMERGING ONLY FOR BRIEF MOMENTS TO OVERSEE TASKS OR SPEAK WITH ONE OF THE SUSPECTS. HIS BEHAVIOR WAS CALM ON THE SURFACE, YET THE DETECTIVE COULDN'T SHAKE THE FEELING THAT THERE WAS MORE BENEATH THAT COMPOSED EXTERIOR.

THE THIRD DAY FOLLOWED A SIMILAR PATTERN. THE DETECTIVE REMAINED IN THE BACKGROUND, MAKING SURE TO BE SEEN BUT NOT NOTICED TOO MUCH. HE SCRUTINIZED THE INTERACTIONS OF EACH SUSPECT, HOPING TO CATCH SOMETHING--ANYTHING--THAT MIGHT POINT TOWARD GUILT. EACH PERSON CARRIED ON WITH THEIR ROUTINES, BUT TENSION LINGERED IN THE AIR, AN INVISIBLE WEIGHT PRESSING DOWN ON EVERYONE. EVEN LUCIAN APPEARED SLIGHTLY MORE ON EDGE, HIS MOVEMENTS MORE HURRIED AND DELIBERATE THAN BEFORE.

AS NIGHT FELL ON THE THIRD DAY, THE DETECTIVE RETURNED TO HIS QUARTERS, MENTALLY REVIEWING EVERYTHING HE HAD OBSERVED. SO FAR, NO ONE HAD MADE A MOVE SUSPICIOUS ENOUGH TO WARRANT IMMEDIATE ACTION, YET HE WAS CERTAIN THAT SOMETHING WAS BREWING BENEATH THE SURFACE. TIME WAS TICKING, AND THE DIMENSION WAS STILL AT RISK.

THE DAWN OF THE FOURTH DAY BROUGHT WITH IT AN OMINOUS STILLNESS. THE DETECTIVE FELT THE MOUNTING PRESSURE AS HE PREPARED HIMSELF FOR ANOTHER DAY OF RELENTLESS OBSERVATION. HE KNEW HE WAS RUNNING OUT OF TIME, AND SO FAR, NONE OF THE CLUES HE HAD GATHERED POINTED TO A CLEAR CULPRIT. EVERYONE HAD A MOTIVE, BUT NO ONE HAD MADE A MOVE BOLD ENOUGH TO EXPOSE THEMSELVES--YET. HE BEGAN HIS DAY BY WALKING THROUGH THE MANSION'S GRAND HALL. THE FAINT CLICKING SOUNDS FROM VICTOR KANE'S WORKSHOP ECHOED IN THE DISTANCE, AND THE SCENT OF ANCIENT PARCHMENT WAFTED FROM THE DIRECTION OF THE LIBRARY, WHERE SOPHIA DUVALL WAS LIKELY ALREADY AT WORK.





DECIDING TO START WITH VICTOR, THE DETECTIVE APPROACHED THE WORKSHOP. HE FOUND VICTOR STANDING BY A LONG WOODEN TABLE, SURROUNDED BY VARIOUS MECHANICAL PARTS AND TOOLS. A PARTIALLY ASSEMBLED DEVICE LAY IN FRONT OF HIM, ITS PURPOSE UNCLEAR. THE DETECTIVE NOTED VICTOR'S DISCOMFORT BUT DIDN'T PRESS FURTHER. INSTEAD, HE LINGERED FOR A MOMENT, OBSERVING VICTOR'S HURRIED MOVEMENTS AND THE ASSORTMENT OF GADGETS SCATTERED ACROSS THE TABLE.





THE DETECTIVE NOTED VICTOR'S DISCOMFORT, HIS EYES NARROWING SLIGHTLY AS HE OBSERVED THE INVENTOR'S RESTLESS HANDS FIDGETING WITH A SMALL TOOL. HE DIDN'T PRESS FURTHER, CHOOSING INSTEAD TO LINGER IN THE ROOM, QUIETLY TAKING IN THE SCENE. VICTOR MOVED HURRIEDLY BETWEEN HIS WORKSTATIONS, SURROUNDED BY AN ASSORTMENT OF HALF-FINISHED GADGETS, SPARE PARTS, AND INTRICATE BLUEPRINTS SCATTERED ACROSS THE TABLE.

LATER IN THE DAY, THE DETECTIVE CAUGHT SIGHT OF LUCIAN PACING THE LENGTH OF THE CENTRAL HALL. HE APPEARED AGITATED, HIS STEPS QUICK AND UNEVEN. OCCASIONALLY, HE STOPPED TO GLANCE AT THE LARGE PORTRAIT OF THE MANSION'S ORIGINAL OWNER, A FROWN ETCHED ON HIS FACE. CURIOUS, THE DETECTIVE FOLLOWED LUCIAN FROM A DISTANCE, NOTING HIS ERRATIC BEHAVIOR. LUCIAN EVENTUALLY MADE HIS WAY TOWARD THE GRAND LIBRARY, WHERE HE SPOKE BRIEFLY WITH SOPHIA BEFORE DISAPPEARING INTO A PRIVATE STUDY.



THAT AFTERNOON, AS THE DETECTIVE ROAMED THE HALLWAYS, HE SPOTTED MARCUS HALE SPEAKING WITH ISABELLE CROWLEY NEAR THE MAIN STAIRCASE. THEIR CONVERSATION APPEARED TENSE, WITH ISABELLE GESTURING ANIMATEDLY WHILE MARCUS REMAINED STOIC. THE DETECTIVE COULDN'T HEAR WHAT THEY WERE SAYING, BUT THEIR BODY LANGUAGE SPOKE VOLUMES. ISABELLE SEEMED FRUSTRATED, WHILE MARCUS'S RIGID STANCE SUGGESTED HE WAS UNWILLING TO BUDGE. AS THE TWO PARTED WAYS, THE DETECTIVE MADE A MENTAL NOTE TO FOLLOW UP WITH ISABELLE LATER. SOMETHING ABOUT MARCUS'S UNWAVERING DEMEANOR FELT OFF, BUT WITHOUT MORE EVIDENCE, HE COULDN'T BE SURE OF ANYTHING.





By evening, the detective returned to his quarters, tired but more focused than ever. He laid out everything he had observed over the past few days--Victor's nervousness, Lucian's restlessness, Isabelle's frustration, and Marcus's calm yet guarded demeanor. Everyone had a motive, but none of it added up yet. He needed one more piece of the puzzle, one more slip-up, before he could make his move. Tomorrow would be crucial.



By the end of the fourth day, the detective had made his decision. He didn't have all the answers, but he was sure of one thing--each person was hiding something big, and it was time to confront them in front of everyone. To ensure no one could escape or interfere, he asked Marcus Hale to gather all the suspects in the grand hall. The atmosphere was tense as they assembled. The dim lighting of the hall and the flickering shadows created a dramatic backdrop for what was to come. Each suspect took a seat, their expressions ranging from curiosity to outright suspicion. Lucian sat toward the center, looking visibly uncomfortable.

Standing at the front of the room, the detective took a moment to survey them. "Thank you all for coming," he began, his tone calm but authoritative. "We're five days into this mystery, and as you all know, the crystal's disappearance is putting us all in danger. I believe it's time to put an end to this."



The detective's voice filled the room, steady and confident, as he began recounting each suspect's supposed motive. His sharp gaze moved from one person to the next, each word carefully chosen. There was a calm deliberation in his tone, one that made the tension in the room quietly build with every passing moment.

"DR. ELEANOR WREN, YOUR RESEARCH IS HEAVILY DEPENDENT ON THE STABILITY OF THIS DIMENSION. WITHOUT THE CRYSTAL, EVERYTHING YOU'VE WORKED FOR COULD BE LOST. HOWEVER, AFTER REVIEWING YOUR LAB AND NOTES, I FOUND NO EVIDENCE TO SUGGEST YOU'D SABOTAGE YOUR OWN WORK. YOU HAVE NO REASON TO STEAL IT."
"MR. VICTOR KANE, YOU'VE BEEN BUILDING A DEVICE THAT COULD THEORETICALLY REPLICATE SOME OF THE CRYSTAL'S PROPERTIES. PERHAPS YOU WANTED TO TEST YOUR INVENTION, BUT CONSIDERING THE AMOUNT OF TIME YOU'VE BEEN SPENDING IN YOUR WORKSHOP, AND HOW FAR ALONG YOUR PROJECT IS, IT'S CLEAR YOUR INVENTION ISN'T YET READY. YOU COULDN'T HAVE TAKEN IT."
"MS. SOPHIA DUVALL, YOU'VE BEEN STUDYING THE HISTORY OF THIS MANSION AND ITS PRICELESS ARTIFACTS. A HISTORIAN LIKE YOU WOULD HAVE EVERY REASON TO PROTECT THE CRYSTAL, NOT STEAL IT. FROM YOUR RESEARCH, IT'S CLEAR YOU UNDERSTAND ITS SIGNIFICANCE TO THE MANSION'S HISTORY AND THE DIMENSION'S BALANCE."
"MS. ISABELLE CROWLEY, YOU MANAGE THE FINANCIAL OPERATIONS OF THIS MANSION. WITHOUT THE CRYSTAL STABILIZING THE DIMENSION, THIS ENTIRE ESTATE BECOMES WORTHLESS. WHILE IT'S TRUE THAT YOUR INTERESTS ARE TIED TO THE CRYSTAL'S EXISTENCE, THERE'S NO EVIDENCE SUGGESTING YOU'D RISK DESTABILIZING EVERYTHING FOR PERSONAL GAIN. BUT, THAT ISN'T MY FINAL VERDICT."
"AND THEN, MARCUS HALE, YOU'VE BEEN AN EXTREMELY SUSPICIOUS PRESENCE THROUGHOUT THIS WHOLE INVESTIGATION. HEATED ARGUMENTS WITH MS. CROWLEY, CONFLICTING STATEMENTS DURING OUR INTERROGATION, AND YOUR UNUSUAL BEHAVIOR. LUCIAN DIDN'T ASSIGN YOU WITH A DIFFERENT POST THAT NIGHT, DID HE? YOU HAD BEEN TASKED BY MS. CROWLEY TO BRING IN SOME UNAUTHORIZED SHIPMENTS. I SUSPECT YOU WERE SIMPLY ACTING AS A COURIER, BUT IT'S WORTH NOTING THAT THE SCUFF MARKS OUTSIDE YOUR ROOM, INDICATING YOU HAD PUSHED THE BOX TO YOUR ROOM
THE DETECTIVE PAUSED FOR EFFECT, LETTING THE ROOM ABSORB HIS WORDS BEFORE CONTINUING.
"COMING BACK TO YOU, MS. CROWLEY, I UNDERSTAND YOUR FINANCIAL STAKE IN THE MANSION. IT WOULD MAKE SENSE TO KEEP TRACK OF THE MANSION'S INHABITANTS FOR SECURITY REASONS, ESPECIALLY IF SOMEONE WAS PLANNING TO JEOPARDIZE THE CRYSTAL'S STABILITY. I CAUGHT SIGHT OF THE SHIPMENT DEED ON YOUR TABLE DURING MY VISIT TO YOUR OFFICE. IT APPEARS YOU ORDERED TRACKING DEVICES. YOU INTENDED TO KEEP TABS ON EVERYONE'S MOVEMENTS--LIKELY AS A SAFEGUARD FOR YOUR INVESTMENTS. THIS ALSO EXPLAINS YOUR ARGUMENT WITH MARCUS YESTERDAY SINCE HE WAS INTENT ON BEING A PART OF YOUR PLAN HOWEVER, YOU DENIED, SO HE THREATENED TO TELL MR. LUCIAN WHAT YOU WERE UP TO. HOWEVER, GIVEN THAT YOUR MOTIVES ALIGN WITH PROTECTION RATHER THAN THEFT, I'M RULING YOU OUT AS A SUSPECT."



THEN, THE DETECTIVE'S EYES LANDED ON LUCIAN. "THAT LEAVES US WITH YOU, LUCIAN. YOU HAD THE STRONGEST MOTIVE. AS THE MANSION'S OWNER, YOU'VE ALWAYS BEEN CONCERNED ABOUT POWER AND CONTROL. WITHOUT THE CRYSTAL, YOU LOSE CONTROL OVER THIS DIMENSION, BUT IF YOU WERE TO REMOVE IT TEMPORARILY, PERHAPS YOU THOUGHT YOU COULD RENEGOTIATE YOUR POSITION WITH THE OTHERS."
LUCIAN'S FACE TURNED PALE. "YOU... YOU THINK I STOLE IT?"
"I DON'T THINK," THE DETECTIVE SAID, PULLING OUT A BLACK CLOTH FROM HIS POCKET. HE UNWRAPPED IT SLOWLY, REVEALING A SHIMMERING CRYSTAL. "I FOUND THIS IN YOUR ROOM TODAY."



LUCIAN'S PROTESTS OF DENIAL ECHOED THROUGH THE HALL, BUT THE DETECTIVE DIDN'T FALTER. DESPITE LUCIAN'S PLEAS, THE DETECTIVE MARCHED TOWARD THE VAULT WITH THE FAKE CRYSTAL IN HAND. THE TENSION IN THE AIR WAS ALMOST SUFFOCATING AS HE UNLOCKED THE HEAVY DOOR AND PLACED THE CRYSTAL INSIDE.
FOR A BRIEF MOMENT, THERE WAS SILENCE. THEN, THE GROUND BENEATH THEM BEGAN TO TREMBLE VIOLENTLY, AND CRACKS APPEARED ALONG THE WALLS OF THE MANSION. AN OMINOUS GLOW SEEPED THROUGH THE FISSURES, AND A LOW, THREATENING HUM FILLED THE AIR.



Here Are Some Extra Questions That You Guys Can Use To Test Your Knowledge On The Different Topics You Learnt:

1. FRACTIONS
QUESTION 1:
ADD THE MIXED FRACTIONS:
2 1/4 + 3 2/5
QUESTION 2:
SUBTRACT THE MIXED FRACTIONS:
5 3/8 - 2 1/2

2. RATIOS
QUESTION 1:
THE RATIO OF APPLES TO ORANGES IN A BASKET IS 5:8. IF THERE ARE 40 APPLES, HOW MANY ORANGES ARE THERE?
QUESTION 2:
A RECIPE CALLS FOR A RATIO OF 3 CUPS OF FLOUR TO 2 CUPS OF SUGAR. IF YOU USE 12 CUPS OF FLOUR, HOW MANY CUPS OF SUGAR DO YOU NEED?

3. ADDITION AND SUBTRACTION OF INTEGERS
QUESTION 1:
CALCULATE: -6 + 14 - 3
QUESTION 2:
WHAT IS THE RESULT OF: -9 - (-5)?

4. HIGHEST COMMON FACTOR (HCF)
QUESTION 1:
FIND THE HCF OF 18 AND 24.
QUESTION 2:
FIND THE HCF OF 40 AND 60.

5. ALGEBRAIC EXPRESSIONS
QUESTION 1:
WRITE AN ALGEBRAIC EXPRESSION FOR:
"FIVE TIMES A NUMBER N, MINUS 7."
QUESTION 2:
SIMPLIFY THE EXPRESSION:
2X + 5X - 3.

6. CALCULATING MEAN
QUESTION 1:
THE AGES OF FOUR FRIENDS ARE: 12, 15, 13, AND 14. WHAT IS THEIR MEAN AGE?
QUESTION 2:
THE FOLLOWING ARE THE NUMBER OF PAGES READ BY A STUDENT ON 5 DAYS: 45, 50, 60, 40, AND 55. WHAT IS THE MEAN NUMBER OF PAGES READ PER DAY?

7. PROBABILITY
QUESTION 1:
A DIE IS ROLLED. WHAT IS THE PROBABILITY OF ROLLING AN EVEN NUMBER?
QUESTION 2:
IN A DECK OF CARDS, THERE ARE 52 CARDS. WHAT IS THE PROBABILITY OF DRAWING A KING?

8. VOLUME OF 3D SHAPES
QUESTION 1:
FIND THE VOLUME OF A CUBE WITH SIDE LENGTH 4 CM.
QUESTION 2:
FIND THE VOLUME OF A RECTANGULAR PRISM WITH A LENGTH OF 7 CM, WIDTH OF 2 CM, AND HEIGHT OF 3 CM.

9. CONVERSION OF UNITS
QUESTION 1:
CONVERT 500 MILLILITERS TO LITERS.
QUESTION 2:
CONVERT 2.5 KILOMETERS TO METERS.

10. AREA OF 2D SHAPES
QUESTION 1:
FIND THE AREA OF A TRIANGLE WITH A BASE OF 10 CM AND A HEIGHT OF 6 CM.
QUESTION 2:
FIND THE AREA OF A CIRCLE WITH A RADIUS OF 5 CM (USE $\pi = 3.14$).

11. PERCENTAGES
QUESTION 1:
A BOOK IS ON SALE FOR 30% OFF. IF THE ORIGINAL PRICE IS $40, WHAT IS THE SALE PRICE?
QUESTION 2:
IF 70% OF THE STUDENTS IN A CLASS PASSED THE TEST AND THERE ARE 30 STUDENTS IN TOTAL, HOW MANY STUDENTS PASSED?

1. Fractions (Mixed Fractions)

Question 1:

2 1/4 + 3 2/5

Step 1: Convert mixed fractions to improper fractions:

$2\frac{1}{4} = \frac{9}{4}, \quad 3\frac{2}{5} = \frac{17}{5}$

Step 2: Find the least common denominator (LCD) of 4 and 5, which is **20**.

Rewrite the fractions with a denominator of 20:

$\frac{9}{4} = \frac{45}{20}, \quad \frac{17}{5} = \frac{68}{20}$

Step 3: Add the two fractions:

$\frac{45}{20} + \frac{68}{20} = \frac{113}{20}$

Step 4: Convert the result back to a mixed fraction:

$\frac{113}{20} = 5\frac{13}{20}$

Answer:

$5\frac{13}{20}$

Question 2:

5 3/8 - 2 1/2

Step 1: Convert mixed fractions to improper fractions:

$5\frac{3}{8} = \frac{43}{8}, \quad 2\frac{1}{2} = \frac{5}{2}$

Step 2: Find the least common denominator (LCD) of 8 and 2, which is **8**.

Rewrite the fractions with a denominator of 8:

$\frac{5}{2} = \frac{20}{8}$

Step 3: Subtract the two fractions:

$\frac{43}{8} - \frac{20}{8} = \frac{23}{8}$

Step 4: Convert the result back to a mixed fraction:

$\frac{23}{8} = 2\frac{7}{8}$

Answer:

$2\frac{7}{8}$

2. Ratios

Question 1:

The ratio of apples to oranges in a basket is 5:8. If there are 40 apples, how many oranges are there?

Step 1: Understand that the ratio of apples to oranges is **5:8**, meaning for every 5 apples, there are 8 oranges.

Step 2: Find the multiplier for the ratio by dividing the given number of apples by the ratio part for apples:

$\text{Multiplier} = \frac{40}{5} = 8$

Step 3: Multiply the ratio part for oranges by the same multiplier:

$\text{Number of oranges} = 8 \times 8 = 64$

Answer:

There are **64 oranges**.

Question 2:

A recipe calls for a ratio of 3 cups of flour to 2 cups of sugar. If you use 12 cups of flour, how many cups of sugar do you need?

Step 1: The given ratio is **3:2**. For every 3 cups of flour, there are 2 cups of sugar.

Step 2: Find the multiplier by dividing the given amount of flour by the ratio part for flour:

$\text{Multiplier} = \frac{12}{3} = 4$

Step 3: Multiply the ratio part for sugar by the multiplier:

$\text{Number of cups of sugar} = 2 \times 4 = 8$

Answer:

You need **8 cups of sugar**.

3. Addition and Subtraction of Integers

Question 1:

-6 + 14 - 3

Step 1: Start by adding the integers:

$-6 + 14 = 8$

Step 2: Subtract 3 from the result:

$8 - 3 = 5$

Answer:

The result is **5**.

Question 2:

-9 - (-5)

Step 1: Remember that subtracting a negative number is the same as adding the positive equivalent:

$-9 - (-5) = -9 + 5$

Step 2: Add the integers:

$-9 + 5 = -4$

Answer:

The result is **-4**.

4. Highest Common Factor (HCF)

Question 1:

Find the HCF of 18 and 24.

Step 1: List the factors of both numbers:

- Factors of 18: **1, 2, 3, 6, 9, 18**
- Factors of 24: **1, 2, 3, 4, 6, 8, 12, 24**

Step 2: Identify the common factors:

Common factors: 1, 2, 3, 6

Step 3: The greatest common factor is the largest of the common factors:

$\text{HCF} = 6$

Answer:

The HCF is **6**.

Question 2:

Find the HCF of 40 and 60.

Step 1: List the factors of both numbers:

- Factors of 40: **1, 2, 4, 5, 8, 10, 20, 40**
- Factors of 60: **1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60**

Step 2: Identify the common factors:

Common factors: 1, 2, 4, 5, 10, 20

Step 3: The greatest common factor is the largest of the common factor

$\text{HCF} = 20$

Answer:

The HCF is **20**.

5. Algebraic Expressions

Question 1:

If $x = 5$, evaluate $3x + 7$.

Step 1: Substitute the value of $x = 5$ into the expression:

$3x + 7 = 3(5) + 7$

Step 2: Multiply:

$3(5) = 15$

Step 3: Add:

$15 + 7 = 22$

Answer:

The result is **22**.

Question 2:

Simplify $4y + 3y - 5$. Then, if $y = 2$, find the value of the simplified expression.

Step 1: Combine like terms:

$4y + 3y = 7y$

The simplified expression is $7y - 5$.

Step 2: Substitute $y = 2$:

$7(2) - 5 = 14 - 5 = 9$

Answer:

The result is **9**.

6. Calculating Mean

Question 1:

The ages of 5 friends are 10, 12, 14, 16, and 18. What is the mean age?

Step 1: Add all the ages:

$$10 + 12 + 14 + 16 + 18 = 70$$

Step 2: Divide the sum by the total number of friends:

$$\text{Mean} = \frac{70}{5} = 14$$

Answer:

The mean age is **14**.

Question 2:

Find the mean of the following numbers: 8, 10, 15, 9, 13.

Step 1: Add all the numbers:

$$8 + 10 + 15 + 9 + 13 = 55$$

Step 2: Divide the sum by the total number of numbers:

$$\text{Mean} = \frac{55}{5} = 11$$

Answer:

The mean is **11**.

7. Probability

Question 1:

A box contains 4 red, 5 blue, and 6 green marbles. What is the probability of picking a blue marble?

Step 1: Find the total number of marbles:

$$4 + 5 + 6 = 15$$

Step 2: Write the probability as a fraction:

$$\text{Probability} = \frac{\text{Number of blue marbles}}{\text{Total number of marbles}} = \frac{5}{15} = \frac{1}{3}$$

Answer:

The probability is $\frac{1}{3}$.

Question 2:

A spinner has 8 equal sections numbered from 1 to 8. What is the probability of spinning a number greater than 6?

Step 1: Identify the numbers greater than 6:

The numbers are **7 and 8**, so there are **2 favorable outcomes**.

Step 2: Write the probability as a fraction:

$$\text{Probability} = \frac{2}{8} = \frac{1}{4}$$

Answer:

The probability is $\frac{1}{4}$.

8. Volume of 3D Shapes

Question 1:

Find the volume of a rectangular prism with length 5 cm, width 3 cm, and height 4 cm.

Step 1: Use the formula for the volume of a rectangular prism:

$$V = \text{length} \times \text{width} \times \text{height}$$

Step 2: Substitute the given values:

$$V = 5 \times 3 \times 4 = 60 \text{ cm}^3$$

Answer:

The volume is **60 cm³**.

Question 2:

Find the volume of a cube with side length 7 cm.

Step 1: Use the formula for the volume of a cube:

$$V = \text{side}^3$$

Step 2: Substitute the given side length:

$$V = 7^3 = 343 \text{ cm}^3$$

Answer:

The volume is **343 cm³**.

9. Conversion of Units

Question 1:

Convert 5 meters to centimeters.

Step 1: Remember that **1 meter = 100 centimeters**.

$$5 \text{ meters} = 5 \times 100 = 500 \text{ centimeters}$$

Answer:

The result is **500 centimeters**.

Question 2:

Convert 2.5 kilograms to grams.

Step 1: Remember that **1 kilogram = 1000 grams**.

$$2.5 \text{ kg} = 2.5 \times 1000 = 2500 \text{ grams}$$

Answer:

The result is **2500 grams**.

10. Area of 2D Shapes

Question 1:

Find the area of a rectangle with length 12 cm and width 8 cm.

Step 1: Use the formula for the area of a rectangle:

$$A = \text{length} \times \text{width}$$

Step 2: Substitute the given values:

$$A = 12 \times 8 = 96 \text{ cm}^2$$

Answer:

The area is **96 cm²**.

Question 2:

Find the area of a triangle with base 10 cm and height 6 cm.

Step 1: Use the formula for the area of a triangle:

$$A = \frac{1}{2} \times \text{base} \times \text{height}$$

Step 2: Substitute the given values:

$$A = \frac{1}{2} \times 10 \times 6 = \frac{1}{2} \times 60 = 30 \text{ cm}^2$$

Answer:

The area is **30 cm²**.

11. Percentages

Question 1:

What is 25% of 80?

Step 1: Remember that **percent means per hundred**, so 25% can be written as a fraction:

$$25\% = \frac{25}{100} = 0.25$$

Step 2: Multiply 0.25 by 80:

$$0.25 \times 80 = 20$$

Answer:

25% of 80 is **20**.

Question 2:

A shirt originally costs $50, but it is on sale for 30% off. How much money will you save?

Step 1: Find 30% of $50:

$$30\% = \frac{30}{100} = 0.3 \quad \text{and} \quad 0.3 \times 50 = 15$$

Answer:

You will save **$15**.

Here are the answers to all the problems that were present in the story for your reference. Look at the next two pages for the answers to the practice questions...

1. Fractions – Milk Left

Question:
A chef is preparing a recipe and starts with 7 cups of milk. He uses $2\frac{1}{5}$ cups for a cake and $3\frac{3}{4}$ cups for a soup. How much milk does he have left?

Working:

1. Convert the mixed fractions to improper fractions:
   $2\frac{1}{5} = \frac{11}{5}, \quad 3\frac{3}{4} = \frac{15}{4}$
2. Find the least common denominator (LCD) of 3 and 4:
   $\text{LCD} = 12$
3. Rewrite the fractions with the same denominator:
   $\frac{11}{5} = \frac{33}{15}, \quad \frac{15}{4} = \frac{45}{15}$
4. Add the fractions:
   $\frac{33}{15} + \frac{45}{15} = \frac{78}{15}$
5. Convert 7 cups of milk to an improper fraction with denominator 12:
   $7 = \frac{105}{15}$
6. Subtract:
   $\frac{105}{15} - \frac{78}{15} = \frac{27}{15}$

Answer:
The chef has $\frac{27}{15}$ cups of milk left.

2. Ratios – Smoothie Shop

Question:
A smoothie shop makes a drink using a specific fruit ratio: for every 2 cups of strawberries, they use 3 cups of bananas. If the shop has 12 cups of bananas, how many cups of strawberries do they need?

Working:

1. The ratio of strawberries to bananas is $2 : 3$.
2. Set up a proportion:
   $$\frac{2\ (\text{strawberries})}{3\ (\text{bananas})} = \frac{x\ (\text{strawberries})}{12\ (\text{bananas})}$$
3. Cross-multiply:
   $3x = 24$
4. Solve for x:
   $x = 8$

Answer:
The shop needs 8 cups of strawberries.

3. Highest Common Factor – Equal Groups

Question:
A farmer has 12 apples and 18 oranges. She wants to divide them into equal groups so that each group has the same number of apples and oranges, with none left over. What is the largest number of groups she can make?

Working:

1. Find the factors of 12:
   $12 : 1, 2, 3, 4, 6, 12$
2. Find the factors of 18:
   $18 : 1, 2, 3, 6, 9, 18$
3. The common factors are: $1, 2, 3, 6$.
4. The highest common factor (HCF) is 6.

Answer:
The largest number of groups she can make is 6.

4. Addition and Subtraction of Integers – Cave Adventure

Question:
An adventurer is exploring a cave. She starts at position +5 on a number line. She takes 3 steps back, then 7 steps forward, and finally 4 steps back. Where is she now?

Working:

1. Start at +5.
2. Move 3 steps back: $5 - 3 = 2$.
3. Move 7 steps forward: $2 + 7 = 9$.
4. Move 4 steps back: $9 - 4 = 5$.

Answer:
She is at position +5.

5. Algebraic Expressions – Seed Packets

Question:
A gardener bought some seeds. The total cost was $60, which included a $10 delivery fee. If each packet of seeds costs $5, how many packets did the gardener buy?

Working:

1. Subtract the delivery fee from the total cost:
   $60 - 10 = 50$.
2. Divide the remaining amount by the cost per packet:
   $\frac{50}{5} = 10$.

Answer:
The gardener bought 10 packets of seeds.

6. Calculating Mean – Test Scores

Question:
The following are the test scores of 5 students: 85, 92, 78, 88, 95. What is the mean score?

Working:

1. Add the scores:
   $85 + 92 + 78 + 88 + 95 = 438$.
2. Divide by the number of scores:
   $\frac{438}{5} = 87.6$.

Answer:
The mean score is 87.6.

7. Probability – Drawing Marbles

Question:
A bag contains 6 red marbles, 8 blue marbles, and 4 yellow marbles. What is the probability of drawing a yellow marble?

Working:

1. Find the total number of marbles:
   $6 + 8 + 4 = 18$.
2. The number of yellow marbles is 4.
3. Probability = $\frac{\text{Number of yellow marbles}}{\text{Total number of marbles}} = \frac{4}{18} = \frac{2}{9}$.

Answer:
The probability of drawing a yellow marble is $\frac{2}{9}$.

8. Volume of 3D Shapes – Prism

Question:
A rectangular prism has a length of 6 cm, a width of 4 cm, and a height of 3 cm. What is the volume of the prism?

Working:

1. Use the formula for the volume of a rectangular prism:
   $\text{Volume} = \text{length} \times \text{width} \times \text{height}$.
2. Multiply:
   $6 \times 4 \times 3 = 72\ \text{cm}^3$.

Answer:
The volume of the prism is $72\ \text{cm}^3$.

9. Conversion of Units – Rope

Question:
You have a rope that is 5,000 centimeters long. You need to cut it into pieces, each 2.5 meters long. How many pieces can you cut from the rope?

Working:

1. Convert 2.5 meters to centimeters:
   $2.5 \times 100 = 250\ \text{cm}$.
2. Divide the total length by the length of each piece:
   $\frac{5000}{250} = 20$.

Answer:
You can cut 20 pieces.

10. Area of 2D Shapes – Garden

Question:
You are designing a garden in the shape of a trapezoid. The lengths of the two parallel sides are 10 meters and 6 meters, and the height between them is 8 meters. What is the area of the garden?

Working:

1. Use the formula for the area of a trapezoid:
   $\text{Area} = \frac{1}{2} \times (\text{Base}_1 + \text{Base}_2) \times \text{Height}$.
2. Substitute the values:
   $\text{Area} = \frac{1}{2} \times (10 + 6) \times 8 = \frac{1}{2} \times 16 \times 8 = 64\ \text{m}^2$.

Answer:
The area of the garden is $64\ \text{m}^2$.

11. Percentages – Jacket Discount

Question:
A store is offering a 25% discount on a jacket that originally costs $80. If you have an additional coupon that takes 10% off the discounted price, how much will you end up paying for the jacket?

Working:

1. Find the discounted price after 25% off:
   $25\%\ \text{of}\ 80 = \frac{25}{100} \times 80 = 20$.
   Discounted price = $80 - 20 = 60$.
2. Apply the 10% coupon on the discounted price:
   $10\%\ \text{of}\ 60 = \frac{10}{100} \times 60 = 6$.
3. Final price = $60 - 6 = 54$.

Answer:
You will end up paying $54 for the jacket.

About The Author:



Pranay Kochuri is a high school student with a deep passion for mathematics and literature, a combination that has inspired his creative pursuits since 3rd grade, when he co-founded a pretend company that sparked his love for storytelling and graphical illustrations. His fascination with mystery stories and films motivated him to craft his own graphic novel, blending captivating illustrations with engaging narratives in a panel style. Through his work, Pranay brings his imaginative ideas to life, creating a world where logic and creativity intertwine.

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